

CHAPTER II

LITERATUE REVIEW

2.1. Eating Disorders

Many American's family are suffering from eating disorders every year. Most of them are adolescent and teenage girls. The amount are around 5% of total population of teenage girls. [3]

The National Institute of Mental Health states that during adolescent and teenage year women are susceptible to the wrong body image. Many try to diet and keep a slim figure and/or try stagnant dieting. The organization also revealed that certain sports and careers are especially prone to reinforcing the need to keep a figure that society think fit, even it does not matches with the body types.

Eating disorder are hard to be detected before it is reached a late stage of disorder. Usually the person who done an eating disorder activity are doing it in secret and done it with a great deal of Shame. "Binging" and "Purging" are the most common behavior of eating disorder. "Binging" refer to the action of overeating foods (usually junk food) at a very short eating window. "Purging" refer to action to clear of eaten food (usually by forcing to vomit) so that they still can eat without gaining weight.

Eating disorders are serious problems and need to be diagnosed and treated like any medical disease. If they continue to go untreated, these behaviors can result in future severe medical complications that can be life-threatening.

2.1.1. Obesity

Obesity is a medical condition in which excess body fat has accumulated to the extent that it may have a negative effect on health, leading to reduced life expectancy and/or increased health problems. [11]

To know whether a person is getting obesity or not the most usual way is by counting the Body Mass Index (BMI). BMI obtained by dividing a person's weight by the square of the person's height, exceeds 30 kg/m^2 , with the range $25\text{-}30 \text{ kg/m}^2$ defined as overweight. Some East Asian countries use stricter criteria.

Obesity increases the likelihood of various diseases, particularly heart disease, type 2 diabetes, obstructive sleep apnea, certain types of cancer, and osteoarthritis. Obesity is most commonly caused by a combination of excessive food energy intake, lack of physical activity, and genetic susceptibility, although a few cases are caused primarily by genes, endocrine disorders, medications, or psychiatric illness. Evidence to support the view that some obese people eat little yet gain weight due to a slow metabolism is limited. On average, obese people have a greater energy expenditure than their thin counterparts due to the energy required to maintain an increased body mass

Obesity are usually caused by lack of physical activity with the excessive food energy intake, Eventhough there are some people who have a genetic susceptibility to obesity, endocrine disorders, medications, or psychiatric illness, But the percentage are very small compared to population of obese people. Obese

people are having greater risk to have various diseases like heart disease, type 2 diabetes, obstructive sleep apnea, certain types of cancer, and osteoarthritis [8]

The usual treatment for obesity is by dieting. Dieting is to choose, limit, and select the kind of food to be consume. Usually to improve the quality of a diet plan obese patient is ordered to reduce the consumption of energy-dense foods, some of the foods are foods with high fat and sugars. Not just controlling the energy intake it also usually suggested for the patient to practice exercise. Exercise suitable for obese patient is usually exercise with high Intensity with short interval training. Acute obesity patient are also usually take anti-obesity drugs that can reduce appetite or decrease fat absorption. There are also surgery to reduce stomach volume and/or bowel length. This kind of operation is intended to making patients feel full earlier than usual. Thus, lead to smaller portion of food eaten.

Even though it can be avoided easily obesity is still became a major factor of death worldwide. Different with other sickness which is harder to detect an avoided, Obesity is relatively easier to be avoided. [25]

2.1.2. Anorexia nervosa

Anorexia or Anorexia nervosa are an eating disorder that are the opposite of obesity. Usually the symptoms showed are extreme food restriction, odd eating habits or rituals, and an unhealthy obsession of thin figure. Anorexia are usually considered as a Psychiatric disorder because it started with distorted body self-perception, and typically involves excessive weight loss. [1] [21]

Anorexia patient are usually have an unusual fear of gaining weight. The term Anorexia or anorexia nervosa are usually often used interchangeably. The

differences is usually anorexia nervosa patient does not lose their appetites unlike the anorexia patients

People with anorexia nervosa usually have different view of self-image. People with Anorexia nervosa always think that they are not thin enough or became too fat. This kind of self-image make people with anorexia nervosa are always felling hungry but deny themselves all but very small quantities of food. [21]

Anorexia nervosa is 9 times are more usually found in women rather than men. [7].

In 2013 it resulted in about 600 deaths globally up from 400 deaths in 1990. It is a serious health condition with a high incidence of comorbidity and similarly high mortality rate to serious psychiatric disorders. [16]

Up to 3 in 4 people are suffering from depression. In addition, one in every four individuals who are diagnosed with anorexia nervosa also exhibit obsessive-compulsive disorder. [5]

Zucker et al. (2007) proposed that conditions on the autism spectrum make up the cognitive endophenotype underlying anorexia nervosa and appealed for increased interdisciplinary collaboration

2.2. Basal Metabolic Rate (BMR)

Basal metabolic rate (BMR) is the amount of energy expenditure per unit time. It is reported in energy units per unit time ranging from watt (Joule/second) to ml O₂/min or Joule per hour per kg body mass J/(h.kg)). Proper measurement requires a strict set of criteria be met. These criteria include being in a physically

and psychologically undisturbed state, in a thermally neutral environment, while in the post-absorptive state (i.e., not actively digesting food). In Brady metabolic animals, such as fish and reptiles, the equivalent term standard metabolic rate (SMR) is used. It follows the same criteria as BMR, but requires the documentation of the temperature at which the metabolic rate was measured. This makes BMR a variant of standard metabolic rate measurement that excludes the temperature data, a practice that has led to problems in defining "standard" rates of metabolism for many mammals. [11]

BMR estimation formulas

Several prediction equations exist. The first formula are came from Harris-Benedict equation, which was created in 1919.

The Original Harris-Benedict Equation:

- for men, $P = \left(\frac{13.7516m}{1\text{ kg}} + \frac{5.0033h}{1\text{ cm}} - \frac{6.7550a}{1\text{ year}} + 66.4730 \right) \frac{kcal}{day}$
- for women, $P = \left(\frac{9.5634m}{1\text{ kg}} + \frac{1.8496h}{1\text{ cm}} - \frac{4.6756a}{1\text{ year}} + 666.0955 \right) \frac{kcal}{day}$

Where

- P is total heat production at complete rest
- m is mass (kg)
- h is height (cm)
- a is age (years)

In 1984, the original Harris-Benedict equations were revised using new data.

The Revised Harris-Benedict Equation:

- for men, $P = \left(\frac{13.397m}{1\text{ kg}} + \frac{4.799h}{1\text{ cm}} - \frac{5.677a}{1\text{ year}} + 88.362 \right) \frac{\text{kcal}}{\text{day}}$
- for women, $P = \left(\frac{9.247m}{1\text{ kg}} + \frac{3.098h}{1\text{ cm}} - \frac{4.330a}{1\text{ year}} + 447.593 \right) \frac{\text{kcal}}{\text{day}}$

It was the best prediction equation until 1990, when Mifflin, MD; St Jeor, ST; Hill, LA; Scott, BJ; Daugherty, SA; Koh, YO. Introduced new equation:

The Mifflin St Jeor Equation:

$$P = \left(\frac{10.0m}{1\text{ kg}} + \frac{6.25h}{1\text{ cm}} - \frac{5.0a}{1\text{ year}} + s \right) \frac{\text{kcal}}{\text{day}}$$

Where

- s is +5 for males and -161 for females[23].

2.3. Dieting

One way to cure eating disorder is by went on a diet. Dieting is a practice of eating food following a certain pattern to alter the body weight and eating habit/ritual. People who undergo a diet plan is usually advised to take a physical exercise in order to lose, gain (usually in the form of muscle) and maintaining body weight [24].

There are 4 widely recognized kinds of diet:

2.3.1. Low-fat diets

A low-fat diet is one that restricts fat and often saturated fat and cholesterol as well. Low-fat diets are intended to reduce diseases such as heart disease and obesity. Reducing fat in the diet can make it easier to cut calories. Fat provides nine

calories per gram while carbohydrates and protein each provide four calories per gram, so choosing low-fat foods makes it possible to eat a larger volume of food for the same number of calories. The Institute of Medicine recommends limiting fat intake to 35% of total calories to help prevent obesity and to help control saturated fat intake [24].

2.3.2. Low-carbohydrate diets

Low-carbohydrate diets or low-carb diets are dietary programs that restrict carbohydrate consumption, often for the treatment of obesity or diabetes. Foods high in easily digestible carbohydrates (e.g., sugar, bread, pasta) are limited or replaced with foods containing a higher percentage of fats and moderate protein (e.g., meat, poultry, fish, shellfish, eggs, cheese, nuts, and seeds) and other foods low in carbohydrates (e.g., most salad vegetables such as spinach, kale, chard and collards), although other vegetables and fruits (especially berries) are often allowed. The amount of carbohydrate allowed varies with different low-carbohydrate diets[3].

Such diets are sometimes 'ketogenic' (i.e., they restrict carbohydrate intake sufficiently to cause ketosis). The induction phase of the Atkins diet is ketogenic.

The term "low-carbohydrate diet" is generally applied to diets that restrict carbohydrates to less than 20% of caloric intake, but can also refer to diets that simply restrict or limit carbohydrates to less than recommended proportions (generally less than 45% of total energy coming from carbohydrates) [24].

2.3.3. Detox diets

Detox diets are dietary plans that claim to have detoxifying effects. The general idea suggests that most food contains contaminants: ingredients deemed unnecessary for human life, such as flavor enhancers, food colorings, pesticides, and preservatives. Scientists, dietitians, and doctors, while generally viewing "detox diets" as harmless (unless nutritional deficiency results), often dispute the value and need of "detox diets", due to lack of supporting factual evidence or coherent rationale. In cases where a person suffers from a disease, belief in the efficacy of a detox diet can result in delay or failure to seek effective treatment [24].

Detox diets can involve consuming extremely limited sets of foods (only water or juice, for example, a form of fasting known as juice fasting), eliminating certain foods (such as fats) from the diet, or eliminating processed foods and alleged irritants. Detox diets are often high in fiber. Proponents claim that this causes the body to burn accumulated stored fats, releasing fat-stored "toxins" into the blood, which can then be eliminated through the blood, skin, urine, feces and breath. Proponents claim that things such as an altered body-odor support the notion that detox diets have an effect; James Dillard of Columbia University explains such phenomena indicates the body undergoing ketosis. Although a brief fast of a single day is unlikely to cause harm, prolonged fasting (as recommended by certain detox diets) can have dangerous health consequences or can even be fatal.

2.3.4. Low-calorie diets

Low-calorie or Calorie restriction (CR), or caloric restriction, is a dietary regimen that is based on low calorie intake. "Low" can be defined relative to the subject's previous intake before intentionally restricting calories, or relative to an average

person of similar body type. Calorie restriction without malnutrition has been shown to work in a variety of species, among them yeast, fish, rodents and dogs to decelerate the biological aging process, resulting in longer maintenance of youthful health and an increase in both median and maximum lifespan. The life-extending effect of calorie restriction is not universal. Wild mice, for instance, do not live longer when on a calorie restricted diet [11].

In humans the long-term health effects of moderate CR with sufficient nutrients are unknown.

Two main lifespan studies have been performed involving nonhuman primates (rhesus monkeys). One, begun in 1987 by the National Institute on Aging, published interim results in August 2012 indicating that CR confers health benefits in these animals, but did not demonstrate increased median lifespan; maximum lifespan data are not yet available, as the study is still ongoing. A second study by the University of Wisconsin beginning in 1989 issued preliminary lifespan results in 2009, and final results in 2014. It found that CR primates were only 36.4% as likely to die from age-related causes when compared with control animals, and had only 56.2% the rate of death from any cause[9] [2].

Effects on humans

In humans the long-term health effects of moderate CR with sufficient nutrient are unknown.

Negative effects

Malnutrition may result in serious deleterious effects, as it has been shown in the Minnesota Starvation Experiment. This study was conducted during World War II

on a group of lean men, who restricted their calorie intake by 45% for 6 months, and composed roughly 90% of their diet with carbohydrates. As expected, this malnutrition resulted in many positive metabolic adaptations (e.g. decreased body fat, blood pressure, improved lipid profile, low serum T3 concentration, and decreased resting heart rate and whole-body resting energy expenditure), but also caused a wide range of negative effects, such as anemia, lower extremity edema, muscle wasting, weakness, neurological deficits, dizziness, irritability, lethargy, and depression[21].

Musculoskeletal losses

Short-term studies in humans report loss of muscle mass and strength and reduced bone mineral density.

The authors of a 2007 review of the CR literature warned that *"[i]t is possible that even moderate calorie restriction may be harmful in specific patient populations, such as lean persons who have minimal amounts of body fat."* [22].

Low BMI, high mortality

CR diets typically lead to reduced body weight, yet reduced weight can come from other causes and is not in itself necessarily healthy. In some studies, low body weight has been associated with increased mortality, particularly in late middle-aged or elderly subjects. Low body weight in the elderly can be caused by pathological conditions associated with aging and predisposing to higher mortality (such as cancer, chronic obstructive pulmonary disorder, or depression) or of the cachexia (wasting syndrome) and sarcopenia (loss of muscle mass, structure, and function). One of the more famous of such studies linked a body mass index (BMI)

lower than 18 in women with increased mortality from noncancer, non-cardiovascular disease causes. The authors attempted to adjust for confounding factors (cigarette smoking, failure to exclude pre-existing disease); others argued that the adjustments were inadequate.

"Epidemiologists from the ACS (American Cancer Society), American Heart Association, Harvard School of Public Health, and other organizations raised specific methodologic questions about the recent Centers for Disease Control and Prevention (CDC) study and presented analyses of other data sets. The main concern ... is that it did not adequately account for weight loss from serious illnesses such as cancer and heart disease ... *[and] failed to account adequately for the effect of smoking on weight ... As a result, the Flegal study underestimated the risks from obesity and overestimated the risks of leanness* [17].

Such epidemiological studies of body weight are not about CR as used in anti-aging studies; they are not about caloric intake to begin with, as body weight is influenced by many factors other than energy intake. Moreover, "the quality of the diets consumed by the low-BMI individuals are difficult to assess, and may lack nutrients important to longevity." Typical low-calorie diets rarely provide the high nutrient intakes that are a necessary feature of an anti-aging calorie restriction diet. As well, "The lower-weight individuals in the studies are not CR because their caloric intake reflects their individual ad libitum set-points, and not a reduction from that set-point."

Triggering eating disorders

In those who already suffer from a binge-eating disorder, calorie restriction can precipitate an episode of binge eating, but it does not seem to pose any such risk otherwise.

Young or pregnant

Long-term calorie restriction at a level sufficient for slowing the aging process is generally not recommended in children, adolescents, and young adults (under the age of approximately 21), because this type of diet may interfere with natural physical growth, as has been observed in laboratory animals. In addition, mental development and physical changes to the brain take place in late adolescence and early adulthood that could be negatively affected by severe calorie restriction. Pregnant women and women trying to become pregnant are advised not to practice calorie restriction, because low BMI may result in ovulatory dysfunction (infertility), and underweight mothers are more prone to preterm delivery.

Miscellaneous concerns

It has also been noted that people losing weight on such diets risk developing cold sensitivity, menstrual irregularities, and even infertility and hormonal changes.

History

The first person to promote calorie restriction as a means of prolonging life was Luigi Cornaro, a 15th-century Venetian nobleman who adopted a calorie restricted diet at age 35 to address his failing health. His restricted diet cured all of his ailments in less than a year, and he went on to live to be 102 years old. His book *Discorsi della vita sobria* (Discourses On the Temperate Life) described his

regimen, which centered on the "quantifying principle" of restricting himself to only 350 grams of food daily (including bread, egg yolk, meat, and soup) and 414 milliliters of wine. The book was extremely successful, and *"was a true reconceptualization of old age. As late as the Renaissance it was largely the negative aspects of this phase of life which were emphasized ... Cornaro's method offered the possibility for the first time not only of a long but also a worthwhile life[18]."*

In 1934, Mary Crowell and Clive McCay of Cornell University observed that laboratory rats fed a severely reduced calorie diet while maintaining micronutrient levels resulted in life spans of up to twice as long as otherwise expected. These findings were explored in detail by a series of experiments with mice conducted by Roy Walford and his student Richard Weindruch. In 1986, Weindruch reported that restricting the calorie intake of laboratory mice proportionally increased their life span compared to a group of mice with a normal diet. The calorie-restricted mice also maintained youthful appearances and activity levels longer and showed delays in age-related diseases. The results of the many experiments by Walford and Weindruch were summarized in their book *The Retardation of Aging and Disease by Dietary Restriction* (1988).

The findings have since been accepted and generalized to a range of other animals. Researchers are investigating the possibility of parallel physiological links in non-human and human primates. In the meantime, many people have independently adopted the practice of calorie restriction in some form[21].

Two randomized control studies on the effects of CR in non-human primates have been conducted: the Wisconsin National Primate Research Center and the National Institute on Aging CR monkey studies. In 1989, scientists at University of Wisconsin started a study involving 46 adult male and 30 female rhesus monkeys. The National Institute on Aging CR monkeys study, started in 1987, involves 60 male and 60 female rhesus monkeys. Monkeys in both studies have been randomized with a 1:1 ratio to 30% CR or to a control diet. Results are being periodically published.

A group of scientists at Washington University in St.Louis have studied long-term physiologic, metabolic and molecular effects of CR in a small group of healthy lean men and women.

In May 2007 a multi-center clinical trial called the CALERIE (Comprehensive Assessment of the Long-term Effects of Reducing Energy Intake) study was begun, to examine the effect of 2 years of sustained 25% CR on: a) slowing aging as assessed by proxy indicators and b) protecting against age-related disease processes. 220 healthy volunteers across 3 sites (Tufts University, Pennington Biomedical Research Center, and Washington University School of Medicine) were recruited.

A study at UCSF called "CRONA" was started in December 2010, and studied 28 long-term CR practitioners over a few months. The study was completed on September 20, 2011. As of August 2012 the results had not yet been published.

2.4. Related Work

This section explains several researches or applications which already exist and are related to this research. They provide the same type of applications and related projects. They are as follow:

2.4.1. MyFitnessPal



Figure 2.1 MyFitnessPal logo on Google play

MyFitnessPal is a free smartphone app and website that tracks diet and exercise to determine optimal nutrients and caloric intake for the users' goals. In a Consumer Reports diet rating, MyFitnessPal was rated the best free program (with 83 points) in overall satisfaction, "maintenance, calorie awareness, and food variety". The user may enter the name of the food or scan the barcode to find the item in a large database of over 5 million foods or may select foods from a list of most frequently eaten foods. MyFitnessPal has different exercises and sometimes the same exercise at different intensities or speeds. The user enters the number of minutes spent exercising and the calories spent is calculated. The calories spent on the exercise is added to the total number of calories that the user should eat per day.[13]

MyFitnessPal is a diet application which focuses in Calorie counter and food diary

The differences between this application and this thesis are the target platform of the app. MyFitnessPal support to runs on iOS and Android platforms, while this thesis is runnable in Android platform only.

2.4.2. Nike +



Figure 2.2 Nike + logo on Google play

The Nike+ Sports Kit is an activity tracker device, developed by Nike, Inc., which measures and records the distance and pace of a walk or run. The Nike+ consists of a small transmitter device attached to or embedded in a shoe, which communicates with either the Nike+ Sportband, a receiver plugged into an iPod Nano, or directly with a 2nd Generation iPod Touch (or higher), iPhone 3GS, iPhone 4, iPhone 4S, iPhone 5, iPhone 5C, iPhone 5S, or a Nike+ Sportwatch. If using the iPod or the iPhone, iTunes software can be used to view the walk or run history.

On September 7, 2010, Nike released the Nike+ Running App (originally called Nike+ GPS) on the App Store, which used a tracking engine powered by MotionX that does not require the separate shoe sensor or pedometer. This application works using the accelerometer and GPS of the iPhone and the accelerometer of the iPod Touch, which does not have a GPS chip. Nike+ Running is compatible with the iPhone 6 and iPhone 6 Plus. [12]

Nike + is an application that focusses in running as a media for achieve healthy lifestyle. It doesn't have calorie counter and full of advertising for Nike products

Table 2.1 Comparison table

Indicators	My Fitness Pal	Nike +	Diet Project
Counting BMR	Yes	No	Yes
Counting Calorie from Food	Yes	No	Yes
Counting Calorie From Exercise	No	Yes	Yes
Support multiple kind of food	Yes	No	Yes
Support multiple kind of exercise	No	No	Yes
Promote certain Brand or diet	No	Yes	Yes
Having Social media with other users	Yes	Yes	No